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Enhancing Well Placement Precision with Seismic-Based Drillbit Tracking Method

Yujin Liu, Aramco Beijing Research Center, Aramco Asia, Beijing, China; Pavel Golikov, Ali Dawood, and Ezzedeen Alfataierge, Geophysics Technology Division, EXPEC Advanced Research Center, Saudi Aramco, Dhahran, Saudi Arabia

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Abstract

This paper introduces a novel method for enhancing drillbit tracking using passive seismic recordings. The method leverages surface passive seismic-while-drilling data, cross-correlated with pilot recordings from various locations along the drill string, to accurately focus, interpret, locate, and visualize drillbit positions. This approach offers an independent and precise estimation of well positions. The method interprets these images as probability distribution functions to simultaneously estimate locations and positional uncertainties. The proposed technique addresses the limitations of traditional well placement methods by providing high-resolution drillbit source images and continuous positional updates. This is particularly advantageous in complex geological settings where precise well placement is critical. The method's ability to deliver accurate drillbit locations without being confined to a spatial grid enhances its applicability across various drilling scenarios. Both synthetic and field data tests validate the method's efficacy, demonstrating its robustness and reliability. The results indicate that this seismic-based approach can serve as a complementary technique to traditional magnetic and gyro Measurement While Drilling (MWD) technologies, potentially mitigating cumulative errors and improving overall well placement accuracy.

Introduction

Accurate well placement is crucial for optimizing production in the oil and gas industry [1]. Precise knowledge of the well path is essential to achieving this goal, as it ensures that the well intersects the target reservoir zones effectively, maximizing hydrocarbon recovery and minimizing drilling costs. Currently, well positioning primarily relies on magnetic measurement-while-drilling (MWD) techniques, often supplemented by gyro measurements [2]. These methods involve the use of sensors placed in the bottom hole assembly (BHA) to measure the Earth's magnetic field and gravitational field, providing information on the wellbore's inclination and azimuth.

However, in long horizontal wells, positional uncertainty increases with measured depth (MD) due to the accumulation of errors in the measurements [3]. This uncertainty is typically greater laterally than vertically, which can lead to significant deviations from the planned well path. The cumulative nature of these errors poses a challenge for accurate well placement, particularly in complex geological settings where precise targeting is essential.

Magnetic MWD tools are susceptible to interference from magnetic anomalies in the subsurface, such as those caused by nearby steel casing or magnetic minerals [4]. Gyro measurements, while less affected by magnetic interference, can still accumulate errors over time and are sensitive to vibrations and shocks during drilling [5]. Both methods require periodic recalibration and are dependent on the accuracy of the initial reference measurements. These limitations highlight the need for an independent and accurate method to estimate well positions, especially in extended reach and horizontal drilling scenarios.

Seismic methods for drillbit tracking offer several advantages over traditional MWD techniques [6]. Firstly, they require no modifications to the BHA and are independent of both magnetic and gravitational fields, making them less susceptible to interference and environmental conditions. Secondly, seismic methods can provide real-time feedback on the drillbit position, allowing for immediate adjustments to the drilling trajectory. This real-time capability is particularly valuable in complex geological settings where rapid decision-making is essential [7].

This paper presents a method for focusing, localizing, and visualizing the drillbit position using surface seismic-while-drilling (SWD) data after cross-correlation. The proposed method involves cross-correlating surface geophone data with pilot recordings from the top drive, downhole tools, or other locations along the drill string. This cross-correlation enhances the signal-to-noise ratio (SNR) of the drillbit signal, enabling accurate localization of the drillbit position. Numerical tests on both 3D synthetic and field data demonstrate that the proposed method effectively visualizes drillbit locations and positional uncertainties, serving as a complementary technology to existing well placement techniques.

Methods

Source Focusing

After cross-correlating surface geophone data with time-correlated pilot data, the traveltime of first arrivals corresponds to the traveltime from the drillbit to the geophone [8]. This cross-correlation process enhances the signal-to-noise ratio (SNR) of the drillbit signal, making it easier to identify and isolate the first arrivals. The first arrivals are the initial seismic waves that travel directly from the drillbit to the geophones, and their traveltimes are crucial for accurate drillbit localization.

Consequently, the drillbit source can be focused by backpropagating the geophone data to the subsurface and applying a zero-lag imaging condition. This imaging condition ensures that the backpropagated wavefields are summed at the correct subsurface locations, effectively focusing the energy at the drillbit source location. Mathematically, this process is expressed as:

$$I(\mathbf{m}) = \sum_{R=1}^{N_R} D_R\big(t = T_R(\mathbf{m})\big), \tag{1}$$

where $\mathbf{m}$ represents the image target in three dimensions, $I(\mathbf{m})$ is the focused source image, $D_R(t)$ is the cross-correlated data at geophone R, and N_R is the total number of geophones. $T_R(\mathbf{m})$ denotes the traveltime from the geophone R to the image target $\mathbf{m}$. Given that the image target is in a three-dimensional space, the traveltime for each geophone R is a 3D array pre-computed via an Eikonal solver.

The Eikonal solver is a numerical method used to compute traveltimes in complex velocity models. It solves the Eikonal equation, which describes the relationship between traveltime and the velocity of seismic waves in the subsurface. By using the Eikonal solver, we can accurately calculate the traveltimes from each geophone to the image target, accounting for variations in the subsurface velocity structure.

The backpropagation process involves reversing the direction of the recorded seismic waves and propagating them back into the subsurface model. This is done by using the pre-computed traveltimes to guide the wavefields to their correct positions in the subsurface. The zero-lag imaging condition ensures that the wavefields are summed at the correct time, effectively focusing the energy at the drillbit source location.

Drillbit localization

The maximum of the focused image corresponds to the drillbit source location. However, the accuracy of this method is constrained by the grid step of the location zone, which can limit the precision of the drillbit position estimation. To address this limitation, the focused image is used as an approximation of the maximum likelihood function for the drillbit source location. This approach allows for a more refined estimation of the drillbit coordinates by interpreting the focused image as a probability distribution function.

The drillbit location coordinates are then defined as averages, calculated using the probability distribution function derived from the focused image. This method provides a more accurate estimation of the drillbit location by averaging the values weighted by the probability distribution function. An advantage of using the probability density function is that the location coordinates are not confined to the nodes of a spatial grid, allowing for a continuous representation of the drillbit position.

In addition to estimating the drillbit coordinates, this approach allows for the calculation of location errors as standard deviations. These standard deviations provide a measure of the positional uncertainties in the drillbit location estimation. Positional uncertainties are influenced by the signal-to-noise ratio (SNR) of the source image. Higher SNR leads to lower uncertainties, while lower SNR increases the uncertainties. Vertical uncertainty is more sensitive to noise due to the origin time-depth trade-off, but both horizontal and vertical uncertainties decrease rapidly with increasing SNR.

The drillbit location and positional uncertainties can be visualized using an ellipsoid. The midpoint of the ellipsoid represents the drill location, while the lengths of the three axes correspond to the positional uncertainties in three dimensions. A larger ellipsoid indicates higher positional uncertainties, providing a vivid representation of both drillbit location and positional uncertainties. This visualization technique helps in understanding the accuracy and reliability of the drillbit position estimation, facilitating real-time drilling navigation and decision-making.

By employing this probabilistic approach, the method enhances the precision of drillbit localization and provides a comprehensive understanding of the associated uncertainties. This is crucial for optimizing well placement and ensuring accurate drilling operations, especially in complex geological environments.

Workflow

Figure 1 illustrates the workflow from data preparation to drillbit localization, comprising the following stages:

1. **Data Preparation:** This initial phase involves computing traveltime tables based on the geophone locations and the velocity model. These traveltime tables are then stored on disk for subsequent use in the drillbit focusing process. Additionally, first-break events are isolated from the cross-correlated data, ensuring that the data is ready for the next stage. This step is crucial for setting up accurate and reliable data for the next step of drillbit source focusing.
2. **Source Focusing:** In this stage, the preprocessed first-break events are utilized, and the previously computed traveltime tables are read from disk. The first-break events are then backpropagated and focused to the source location, which corresponds to the drillbit location. This step is crucial for accurately determining the position of the drillbit, ensuring precise localization.
3. **Drillbit Localization:** The final stage involves interpreting the source focusing image as a maximum likelihood function of the drillbit source location. Based on this interpretation, the drillbit location and its positional uncertainties are estimated. This stage ensures that the drillbit's position is accurately

identified, allowing for precise drilling operations and minimizing errors, ultimately enhancing the efficiency and safety of the drilling process.

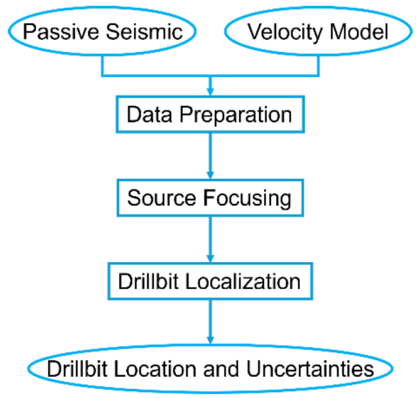


Figure 1—The workflow from data preparation to drillbit localization.

Results

The efficacy of the proposed methodology was validated through comprehensive testing using both 3D synthetic data and real field data. The synthetic data tests involved simulating the drillbit's position and generating seismic responses using finite-difference modeling techniques. These controlled tests allowed for a detailed evaluation of the method's accuracy and robustness under ideal conditions. The field data tests, on the other hand, provided a practical assessment of the methodology's performance in real-world scenarios.

3D Synthetic Data Tests

The proposed method was initially tested on 3D synthetic data, assuming the drillbit is located at a depth of 762 meters. The surface geophones were distributed similarly to the first field trial of seismic-while-drilling [9], as shown in Figure 2(a). The maximum offset is approximately 600 meters, with some geophones missing within this range, leading to a loss of imaging resolution and SNR. The velocity model, shown in Figure 2(b), is also from the field trial. The correlated data was simulated using finite-difference seismic forward modeling with a Ricker wavelet of 15 Hz dominant frequency, as depicted in Figure 3(a).

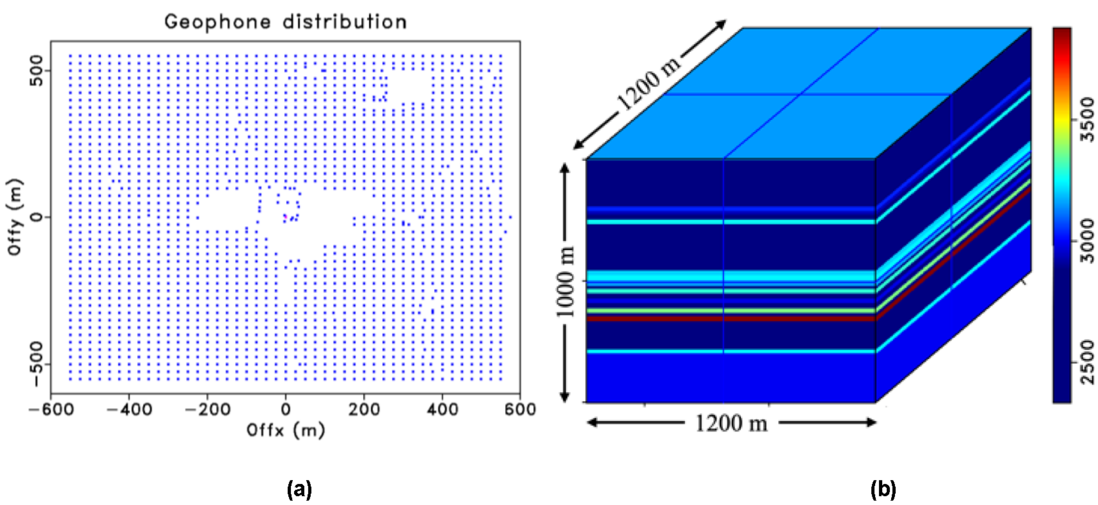


Figure 2—Synthetic data tests. (a) Acquisition geometry; (b) Velocity model.

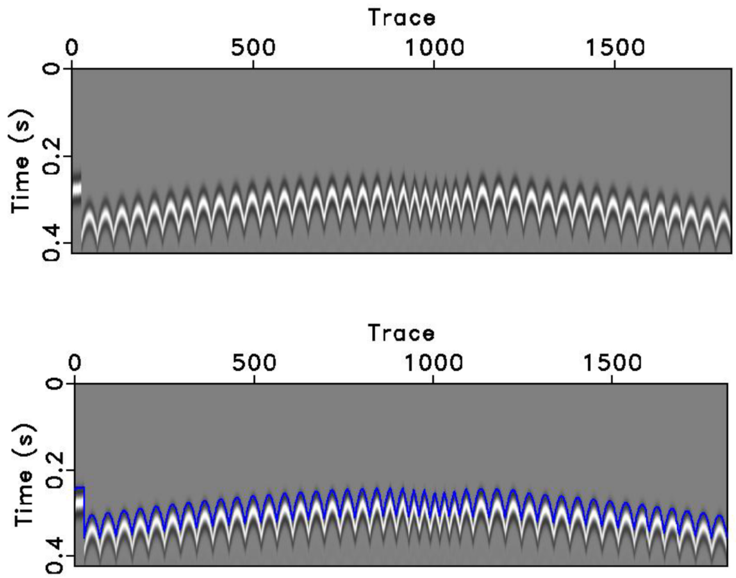


Figure 3—Synthetic data. (a) Synthetic DrillCAM data after correlation; (b) Calculated traveltime overlaid on the synthetic data.

According to the source localization formulas (1) and (2), imaging the drillbit source requires the traveltime from each surface geophone to the subsurface target around the drillbit. This traveltime, a 3D array, is calculated using an Eikonal solver for each surface geophone, with the resulting traveltime volumes stored on disk. These volumes are retrieved during drillbit source localization. To validate the Eikonal solver, we calculated the traveltime from the drillbit to each surface geophone and overlaid these on the synthetic seismic wavefield, as shown in Figure 3(b). The traveltime calculated by the Eikonal solver is reasonably accurate.

After preparing the traveltime tables from each geophone to the image target, we back-propagated the geophone data and applied the zero-lag imaging condition to obtain the drillbit source image. Figure 4 shows the focusing image after applying the focusing imaging condition, which involves stacking all contributions from the geophones. The source image has high resolution and SNR that contains the information of the drillbit position.

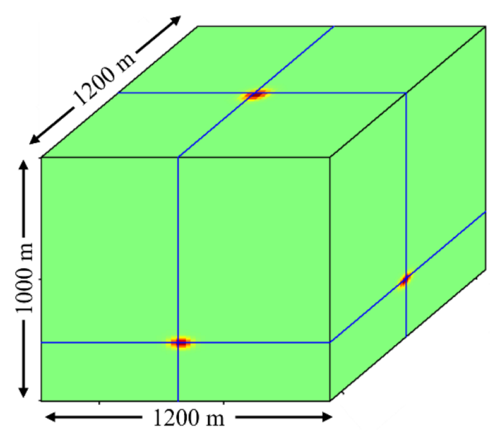
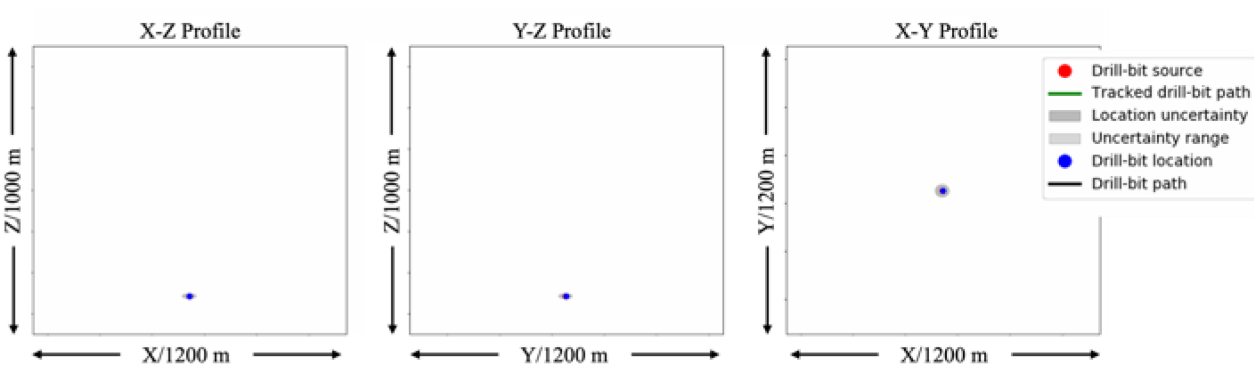


Figure 4—Drillbit source focusing image.

We then visualized the source locations and positional uncertainties by interpreting the calculated source focusing image. Figure 5 shows the visualization results of the focused images. The red dots denote the source locations by focusing the geophone data and the blue dot indicates the true drillbit source location used in the seismic simulation. The grey ellipsoid represents positional uncertainties. The green and black lines denote the drilling path tracked by the drillbit source locations and the drilling path from the drilling logs. Since only single drillbit position is tested here, both lines are invisible in the figure. Most notably, the positional uncertainties have been well estimated in addition to the drillbit location.



(b)

Figure 5—Visualization of drillbit locations and positional uncertainties after interpretation of the source focusing images.

3D Field Data Tests

The proposed method was tested on a 3D field dataset, with drillbit locations sampled at 73 depths ranging from 189.9 meters to 801.3 meters. The distribution of geophones recording the drillbit-generated noise at these depths is shown in Figure 2(a), and the provided 1D interval velocity is shown in Figure 2(b). The seismic data, shown in Figure 6(a), is geophone data after cross-correlation with the pilot data and preprocessing, including noise attenuation and time-shift correlation [10]. Figure 6(b) shows one shot gather corresponding to a drilling depth of 762 meters, with first arrival times picked and stored in the header.

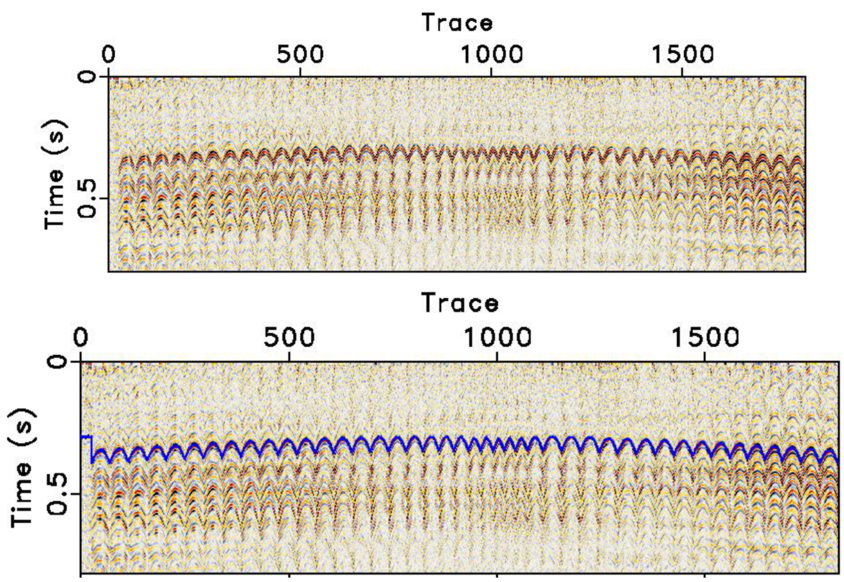


Figure 6—Field data. (a) Field data after correlation; (b) Picked traveltime overlaid on the field data.

Using the velocity model and geophone distribution, we calculated the traveltimes from each geophone to the subsurface image target around the drillbit. To validate the velocity model, we used the Eikonal solver to calculate the traveltime from the drillbit location given in the drilling logs to the surface geophones. Figures 7(a) show both the calculated and picked traveltimes overlaid on the corresponding seismic data, which reveals a good match, leading to accurate drillbit location. Figure 7(b) shows the isolated first breaks by windowing the seismic data around the picked first arrival traveltime.

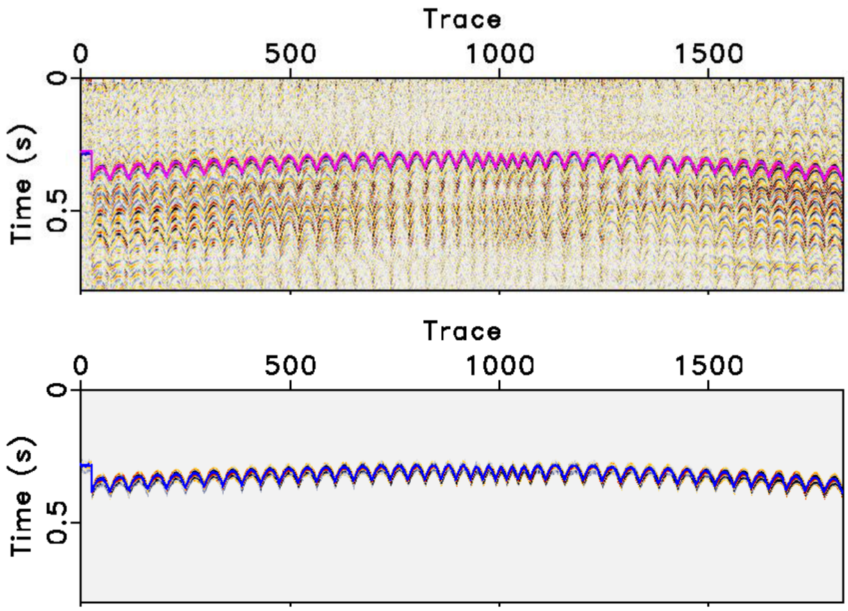


Figure 7—First break extraction. (a) Comparisons of manual picked and calculated traveltimes of first arrivals at the drilling depth of 672 m; (b) Isolated first breaks using the picked first-arrival traveltime.

After preparing the traveltime tables from each geophone to the image target, we back propagated the geophone data and applied focusing imaging condition to obtain the drillbit source image. Figure 8 shows the focusing images after applying the focusing imaging condition at drilling depth of 672 meters. We can observe that the focus image provides crucial information of the drillbit position.

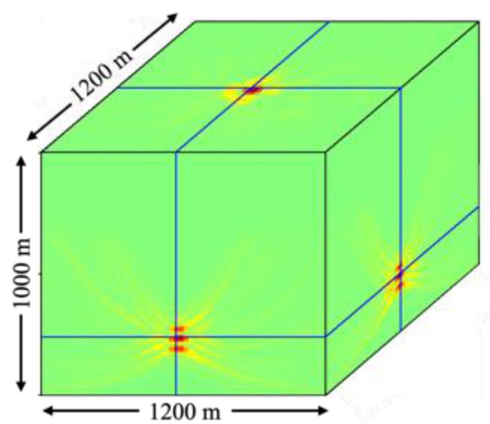
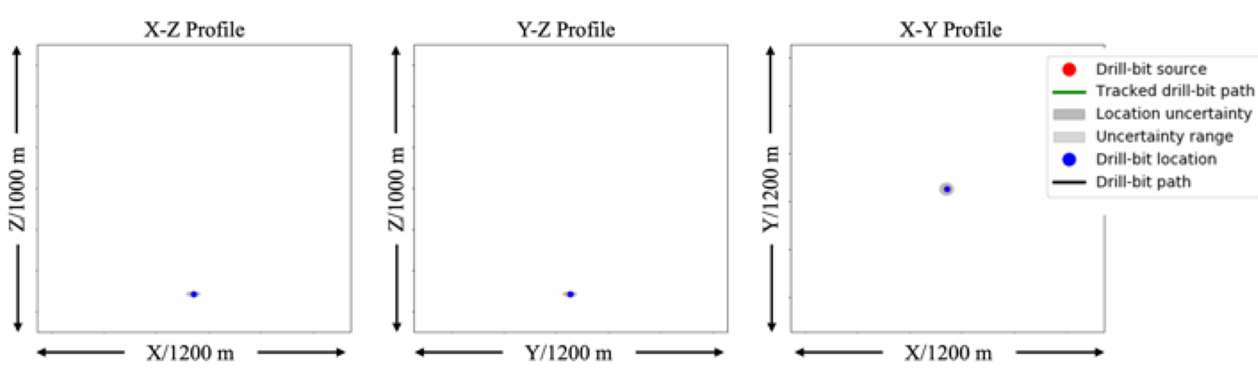


Figure 8—Drillbit source focusing image.

We then visualized the source locations and positional uncertainties by interpreting the calculated source focusing image. Figure 9 shows the visualization results of the focused image. The red dots denote the source locations by focusing the geophone data and the blue dot indicates the true drillbit source location used in the seismic simulation. The grey ellipsoid represents positional uncertainties. Most notably, the positional uncertainties are well estimated in addition to the drillbit location. The green and black lines denote the drilling path tracked by the drillbit source locations and the drilling path from the drilling logs. Since only single drillbit position is tested here, both lines are invisible in the figure.



(b)

Figure 9—Drillbit locations and positional uncertainties after interpretation of the source image.

Discussions

The proposed method significantly enhances the resolution of drillbit source images and provides accurate estimations of drillbit locations and positional uncertainties. The use of probability distribution functions for interpreting the source images allows for precise estimation of drillbit locations without being confined to the nodes of a spatial grid. This probabilistic approach not only improves the precision of drillbit positioning but also offers a continuous representation of the drillbit's trajectory, which is crucial for optimizing well placement in complex geological settings.

However, the method's accuracy is highly dependent on the quality of the legacy velocity model used. Inaccurate velocity models can lead to cumulative drillbit source location errors, particularly as the depth increases. These errors can compromise the reliability of the drillbit positioning, making it essential to refine the velocity model continually. Future work should prioritize the development and integration of more accurate velocity models to enhance the overall precision of drillbit location estimations. Additionally, incorporating real-time updates to the velocity model based on ongoing seismic data could further mitigate positional uncertainties and improve the robustness of the drillbit tracking system. This continuous improvement in the velocity model will be vital for maintaining the high-resolution and accuracy of the proposed method in various drilling scenarios.

An additional potential direction involves the integration of the seismic-based drillbit tracking method with magnetic and gyro Measurement While Drilling (MWD) technologies. By combining these methodologies, we can capitalize on the strengths inherent in each approach. The seismic method's stability in lateral positional accuracy can enhance the real-time data provided by MWD, ensuring precise well placement even in complex geological environments. This hybrid approach has the potential to mitigate the cumulative errors associated with MWD, resulting in a more reliable and accurate drillbit tracking system.

Conclusions

We have successfully developed a novel drillbit tracking method that significantly enhances the precision of well placement. This method leverages passive seismic recordings to deliver high-resolution drillbit source images by backpropagating cross-correlated surface geophone data. The focused source image is interpreted as a probability distribution function of source locations, which can be utilized to estimate drillbit source locations and their positional uncertainties. This probabilistic approach allows for a continuous representation of the drillbit position, eliminating the constraints imposed by grid size and providing high-precision drillbit locations with minimal uncertainties.

The proposed method offers several advantages over conventional drillbit location techniques based on magnetic and gyro MWD. Traditional MWD methods are prone to cumulative errors with increasing measured depth (MD), particularly in long horizontal wells. These errors can lead to significant deviations from the planned well path, compromising the accuracy of well placement. In contrast, the seismic-based technology developed in this study provides an independent and precise estimation of drillbit positions, with positional errors that do not increase with MD. This stability in lateral positional accuracy is crucial for optimizing well placement, especially in complex geological settings where precise targeting is essential.

The efficacy of the proposed method has been validated through comprehensive testing using both 3D synthetic data and real field data. The synthetic data tests demonstrated the method's accuracy and robustness under controlled conditions, while the field data tests provided a practical assessment of its performance in real-world scenarios. The results from both tests confirm that the method can effectively visualize drillbit locations and positional uncertainties, serving as a complementary technology to existing well placement techniques.

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